



Smart IoT **Demand-Side** Load Management

Voltage Stabilization for Solar-Powered Micro-Grids

[COLLEGE PROJECT REVIEW - 8:30 AM]

| Problem Statement



Solar Intermittency

Sudden weather changes (clouds) lead to **Voltage Sags**. Current systems crash because they can't handle heavy loads during these drops.



Grid Instability

Heavy appliances like ACs draw high current during voltage dips, causing a total **Micro-Grid Collapse** or blackout.

"Veyil pogum podhu voltage sudden ah drop aagum... grid crash aagum."

| Project Abstract

This project introduces a **Proactive Demand-Side Management** system using IoT. By utilizing real-time edge computing on an ESP32, the system detects voltage fluctuations in solar micro-grids. When power generation drops, it initiates "**Graceful Degradation**"—automatically shedding non-essential loads (decoration lights, AC) while maintaining power to essential ones (fans, laptops). This prevents total grid failure and ensures **Net-Zero Power Wastage**.

Technical Stack: Hardware

-  **ESP32 WROOM:** Dual-core logic brain with integrated Wi-Fi.
-  **ZMPT101B:** High-precision AC Voltage Sensor.
-  **ACS712:** Current Sensor for load monitoring.
-  **4-Channel Relay:** Automated load-shedding switches.



| Technical Stack: Software



Firmware

C++ via Arduino IDE. Real-time PID-like thresholding logic at the edge.



IoT Dashboard

Blynk Cloud Platform for live visualization and remote manual override.



Communication

ESP-NOW Protocol for ultra-low latency Master-Slave node switching.

| System Workflow



Sense

Read Voltage/Current via
ZMPT & ACS sensors.



Think

ESP32 calculates power gap
vs demand priority.



Act

Relays shed non-essential
loads in <50ms.



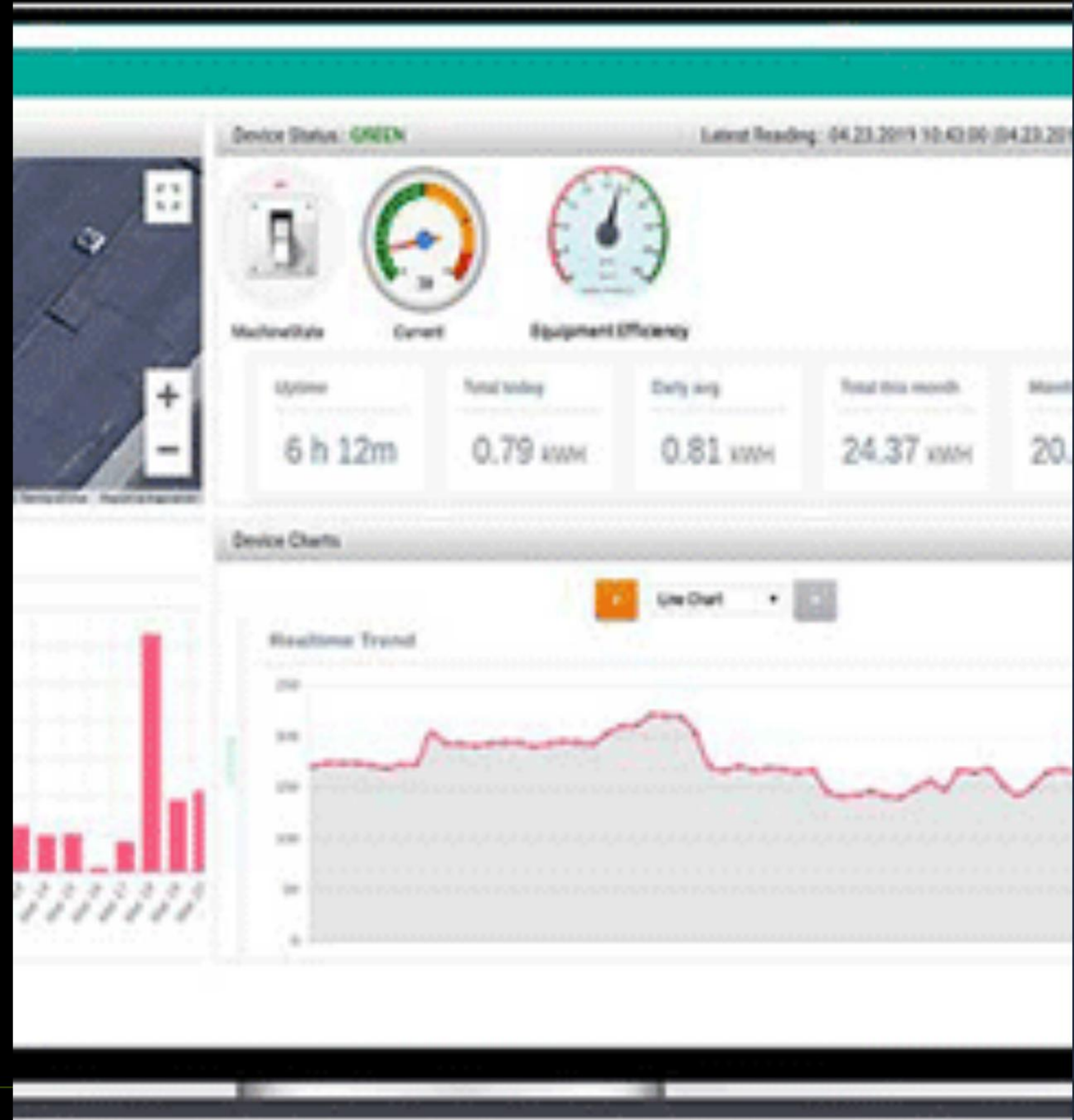
Report

Push live status and alerts to
Blynk IoT app.

Distributed IoT Nodes

Our architecture uses a **Master Node** at the solar inverter and **Wireless Slave Nodes** at every room's switchboard.

This eliminate complex wiring. The Master sends an encrypted wireless command to specific rooms to cut power when solar drops.



| Target Applications



Rural Micro-Grids

Village settlements where EB grid is absent.



Telecom Towers

Critical uptime for remote communication nodes.

600 × 400

Modern Solar Homes

Rooftop solar users in smart cities.

| Novelty & Differentiation

<50ms

Response Speed

Edge processing beats cloud latency.

Proactive vs Reactive: We prevent crashes instead of resetting after failure.

Granular Control: Load-by-load management vs. Whole-house trip.

Economic Scale: Built for ₹3,000 using ESP32 vs. Lakhs for industrial SCADA.

| System Stability Analysis



Stability represents the percentage of cloud-cover events handled without a total grid blackout.

Business & Market Scope

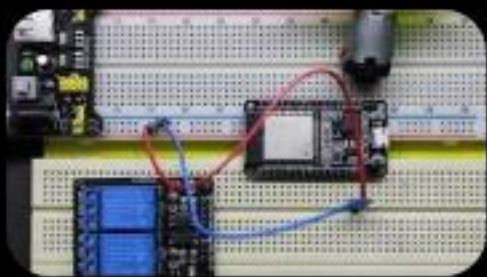
Market Segment	Value Proposition	Scalability
Solar Homeowners	Protects appliances & battery life.	High (Retrofit kit)
Microgrid EPCs	Improves village grid reliability.	B2B Partnerships
Govt. Projects	PM Surya Ghar Yojana integration.	National Impact

Questions?

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SUBMITTED BY: YOUR TEAM NAME
NET-ZERO POWER WASTAGE PROJECT

| Image Sources



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